Biological Networks Part B (Calculation Lab)

Reading: NEWORK PROPERTIES

Nothing in a cell works alone. Genes switch each other on and off, proteins bind to proteins, and metabolites pass through enzymes. In Part A we drew these connections as a graph of nodes (genes, proteins) and edges (interactions). In Part B we measure that graph. Read each property below before starting the calculations.

1. Degree (k) and hubs

Degree asks: how many friends does each person have? It is the number of edges touching a node. In an adjacency matrix it is the row sum. A node with unusually high degree is a **hub**. Removing a hub disrupts many interactions at once, which is why hubs are often essential genes and candidate drug targets.

2. Network density

Density is the fraction of possible edges that actually exist. It tells you how tightly connected the whole system is. E is the number of edges and N the number of nodes.

Tip: in an undirected network the degrees always add up to twice the number of edges, because every edge touches two nodes.

So E = (sum of all degrees) ÷ 2.

3. Path length and diameter

Path length asks: how many handshakes does it take to get a message from one person to another? The **shortest path** between two nodes is the route with the fewest edges, found by breadth-first search (step out one edge at a time from the start node). Short paths mean signals travel quickly, so path length measures communication efficiency.

- **Average path length** = the mean of the shortest-path distances over all pairs of nodes. There are $N(N-1)/2$ pairs in an undirected network.
- **Diameter** = the longest of all the shortest paths.

4. Clustering coefficient (C)

The clustering coefficient asks: asks whether your friends are also friends with each other. A high value means tight local neighborhoods. This is why it is "high in modular biological systems": proteins in the same complex or pathway tend to all interact with one another.

5. Connected components

A connected component is a group of nodes that can all reach each other. If the network splits into islands with no bridge between them, each island is a component. Several islands mean isolated subsystems: separate pathways, or a gap in the data.

Reading: PROPERTIES ANALYSIS TECHNIQUES

1. **Node and Edge Attributes**

Node attributes describe the biological entity itself while edge attributes describe the interaction itself.

2. **Degree Distribution**

Degree tells you about one node. The **degree distribution** tells you about the whole network: of all the nodes, what fraction have 1 connection, what fraction have 2, what fraction have 3, and so on.

3. **Clustering and Modules**

Proteins rarely work alone. They form teams: a protein complex where several subunits bind together, or a pathway where each enzyme hands its product to the next. In a network these teams appear as **modules**: groups of nodes with many connections inside the group and few connections to the rest of the network. Finding modules is how we go from a tangle of edges to a list of functional units.

4. **Centrality Measures**

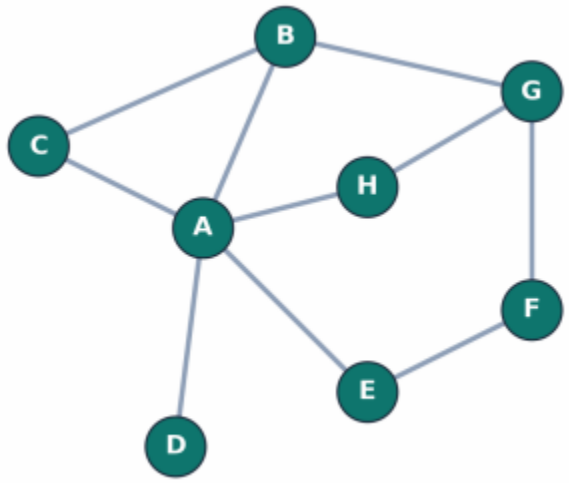
Centrality estimates how important a node is for the connectivity or information flow of the network. In protein interaction networks it is used to find signaling bottlenecks and drug targets.

- **Degree centrality** = the node's degree (number of direct links), often normalized by dividing by $N - 1$.
- **Closeness centrality** estimates how fast information can flow from a node to all others. First find the node's **farness**: the sum of its shortest-path distances to every other node. Closeness is the inverse of farness.
- **Betweenness centrality** measures how often a node lies on the shortest paths between other pairs of nodes. A node with high betweenness is a **bottleneck**: many messages have to pass through it.

Part 2 – The network

The network below comes from slide 3(the notes shared) .

Treat it as a small protein interaction network: each letter is a protein, and each line is an undirected interaction.



Edge list (10 interactions): A–B, A–C, A–D, A–E, A–H, B–C, B–G, E–F, F–G, G–H

Node	Connected to
A	B, C, D, E, H
B	A, C, G
C	A, B
D	A
E	A, F
F	E, G
G	B, F, H
H	A, G

Part 3 – Calculations

Complete every task using the network in Part 2. Write fractions and round decimals to 3 places.

Task 1 – Adjacency matrix

Fill in the matrix: write 1 if the two nodes interact and 0 if they do not. The diagonal is 0, since a protein does not interact with itself here. Check that your matrix is symmetric. Then add a final column for the row sums.

	A	B	C	D	E	F	G	H	Degree
A	0								
B		0							
C			0						
D				0					
E					0				
F						0			
G							0		
H								0	

Task 2 – Degree and hubs

Node	Degree k	Degree Centrality k/(N−1)
A		
B		
C		
D		
E		
F		
G		
H		

1. Add up all the degrees. Use the result to find the number of edges, E.
2. Calculate the average degree of the network.
3. Which node is the hub? Which node is the least connected?
4. Include a column on closeness centrality.

If you have any questions, feel free to reach out*

Submit before the deadline: Attendance is based on submission*

Edit the name of document to include your admission number when submitting *

GOOD DAY!